THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **How Forces Affect Motion** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Newton's first law	Law of inertia.
2	**(a)** are equal in magnitude, opposite in direction and act on different bodies	Third law.
3	**(a)** one-sixth	g_moon is g/6.
4	**(b)** mass	More mass, more inertia.
5	**(a)** act on different bodies	Different bodies.
6	**(c)** increasing the stopping time and reducing the force	F = change in momentum/time.
7	**(c)** newton	N = kg m/s squared.
8	**(b)** the exhaust gases pushed downward exert an equal and opposite force on it	Third law.
9	**(a)** increase the time of stopping and reduce the force	Impulse.
10	**(c)** 1 kg	F = ma.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A body remains at rest or in uniform motion unless acted on by an external force (1); Inertia is the tendency of a body to resist a change in its state (1).	1+1
12	Inertia of motion (1); Body tends to continue to move with the same speed (1).	1+1
13	To increase the time to stop the ball (1); This reduces the force on the hands (1).	1+1
14	Force is the rate of change of momentum (1); F = ma (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Net force = 5 N east (1); a = 5/2 (1); = 2.5 m/s squared (1).	1+1+1
16	a = 3 m/s squared (1); v = at = 9 m/s (1); Starts from rest (1).	1+1+1
17	Foot pushes the ground backward (1); Friction pushes forward (1); On ice friction is low and we slip (1).	1+1+1
	OR: a = -4 m/s squared (1); F = ma (1); = 4000 N opposite to motion (1).	1+1+1
18	Action and reaction act on different bodies (1); Horse pushes the ground backward, ground pushes it forward (1); Net force on the horse-cart system comes from friction of the ground (1).	1+1+1
	OR: p = mv (1); F = m(v - u)/t (1); = ma (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	First law with example of a coin on a card (1); Second law F = ma with a pushed trolley (1); Third law with rocket (1); Statements correct (1); Daily-life relevance (1).	1+1+1+1+1
	OR: Net force 20 N (1); a = 4 m/s squared (1); v = 16 m/s (1); s = 32 m (1); Formulas used (1).	1+1+1+1+1
20	Place a card on a glass with a coin on it (1); Flick the card quickly (1); The coin drops into the glass (1); Pile of coins: strike the bottom coin (1); Inertia of rest (1).	1+1+1+1+1
	OR: Mass is constant (1); Weight = mg (1); On Earth 500 N (1); On Moon about 83 N (1); g on Moon is one-sixth (1).	1+1+1+1+1

Mistakes to avoid:

- Using mass in kg as weight in N (a 5 kg body weighs 50 N when $g = 10$), or saying that the mass of a body changes on the Moon.
- Using the applied force instead of the *net* force in $F = ma$ when friction acts.
- Saying that action and reaction cancel; they are equal and opposite but act on two different bodies.

- Forgetting to convert grams to kilograms (a 20 g bullet is 0.02 kg) in momentum and recoil problems.
- Dropping the negative sign (direction) of recoil velocity, or believing that a heavier gun recoils faster; recoil speed is inversely proportional to the gun's mass.
- Confusing inertia of rest with inertia of motion, or thinking that inertia depends on speed; it depends only on mass.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **How Forces Affect Motion** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** opposite to the direction of motion or tendency of motion	Opposes relative motion.
2	**(d)** stop passengers from moving forward due to inertia	Inertia of motion.
3	**(d)** inertia of rest	Body tends to stay at rest.
4	**(b)** rough surfaces	More irregularities.
5	**(a)** would move forever with constant velocity	No friction.
6	**(c)** inertia of motion	Body tends to keep moving.
7	**(a)** less than sliding friction	Wheels help.
8	**(c)** 5 m/s squared	Net 5 N.
9	**(d)** equal to the rate of change of its momentum	F = dp/dt.
10	**(d)** greater than sliding friction	Needed to start motion.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Mass: quantity of matter, kg, constant (1); Weight: force of gravity, N, varies with place (1).	1+1
12	Friction is very small (1); Foot slips (1).	1+1
13	Every action has an equal and opposite reaction (1); A rocket pushing gas down is pushed up (1).	1+1
14	F = ma (1); = 3000 N (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Burning fuel throws hot gases downward (1); Gases push the rocket upward with an equal and opposite force (1); Works in space too (1).	1+1+1
16	Dust falls off a carpet when beaten (1); A coin on a card drops into a glass when the card is flicked (1); Passengers jerk forward when brakes are applied (1).	1+1+1
17	Impulse = force x time = change in momentum (1); Longer time reduces the force for the same change (1); Avoids injury (1).	1+1+1
	OR: Truck has greater mass and momentum (1); Greater force or time needed (1); More inertia (1).	1+1+1
18	On Earth: 600 N (1); On Moon: 100 N (1); Mass remains 60 kg (1).	1+1+1
	OR: Net force = 20 N (1); a = 20/20 (1); = 1 m/s squared (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	p = mv (1); F proportional to (mv - mu)/t (1); F = ma (k = 1 in SI) (1); F = 3000 N (1); Units (1).	1+1+1+1+1
	OR: Seat belts stop inertia of motion (1); Airbags increase time of stopping (1); Crumple zones absorb energy (1); Headrests prevent whiplash (1); F = change in momentum/time (1).	1+1+1+1+1
20	Causes: surface irregularities (1); Types and sizes (1); Advantages: walking, brakes, grip (1); Disadvantages: wear, heat (1); Methods of control (1).	1+1+1+1+1
	OR: a = 4 m/s squared (1); v = 12 m/s (1); Change in momentum = 60 kg m/s (1); Impulse = 20 x 3 = 60 N s (1); s = 18 m (1).	1+1+1+1+1

Mistakes to avoid:

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- Using the applied force instead of the *net* force in $F = ma$ when friction acts.
- Saying that action and reaction cancel; they are equal and opposite but act on two different bodies.

- Forgetting to convert grams to kilograms (a 20 g bullet is 0.02 kg) in momentum and recoil problems.
- Dropping the negative sign (direction) of recoil velocity, or believing that a heavier gun recoils faster; recoil speed is inversely proportional to the gun's mass.
- Confusing inertia of rest with inertia of motion, or thinking that inertia depends on speed; it depends only on mass.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **How Forces Affect Motion** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** do not change its state of motion	Net force is zero.
2	**(c)** mv	p = mv.
3	**(c)** the gravitational force on it	W = mg.
4	**(c)** zero	No acceleration.
5	**(d)** kg m/s	Mass times velocity.
6	**(d)** 50 N	mg.
7	**(a)** greater acceleration	a = F/m.
8	**(b)** 5 m/s squared	a = F/m.
9	**(c)** the same	Mass is constant.
10	**(d)** N s	Impulse units.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Lubrication (1); Using ball bearings or polishing (1).	1+1
12	Force that gives 1 m/s squared acceleration to 1 kg (1); 1 N = 1 kg m/s squared (1).	1+1
13	Product of mass and velocity (1); kg m/s (1).	1+1
14	Balanced: net force zero (1); Unbalanced: net force not zero, causes acceleration (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	p1 = 20 kg m/s (1); p2 = 15 kg m/s (1); The first has greater momentum (1).	1+1+1
16	Inertia of motion keeps passengers moving forward (1); Belts hold them and spread the force (1); Airbags increase stopping time, reducing force (1).	1+1+1
17	Starting: lower body moves with bus, upper body stays (1); Stopping: lower body stops, upper body continues (1); Inertia of rest and of motion (1).	1+1+1
	OR: Increase: rough surfaces, treads, sand (1); Reduce: oil, bearings, polish (1); Examples used (1).	1+1+1
18	Bullet: 2/0.1 = 20 m/s squared (1); Gun: 2/5 = 0.4 m/s squared (opposite) (1); Same force, opposite directions (1).	1+1+1
	OR: Gravitational force (1); g = 10 m/s squared (9.8) (1); Direction towards the centre of the Earth (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(a) 4 m/s squared (1); (b) 4000 N (1); (c) 50 m (1); (d) 20000 kg m/s (1); Force = change in momentum/time = 4000 N (1).	1+1+1+1+1
	OR: Ball rolls down one slope and rises up the other to the same height (1); Reducing the slope of the second plane makes the ball travel farther (1); On a horizontal smooth plane the ball would go on forever (1); Motion does not need a force (1); Newton's first law (1).	1+1+1+1+1
20	Swimmer pushing water (1); Rocket (1); Gun and bullet (1); Walking (1); Act on different bodies (1).	1+1+1+1+1
	OR: Swimmer pushes water backward (1); Boat's oar pushes water (1); Rocket expels gas (1); Reaction moves the body forward (1); Equal and opposite (1).	1+1+1+1+1

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